
Design and Implementation of a Modular Dual-Evaporator Transcritical High Temperature Heat Pump

Catalogue of Possible Cycle Configurations

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Date: October 5, 2025

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Abstract

This document presents the various possible cycle configurations for the modular high-temperature heat pump (HTHP) system, along with their corresponding schematics and specified boundary conditions.

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Note on AI assistance

This document was proofread with the assistance of **ChatGPT**, an AI language model developed by **OpenAI**.

1 Subcritical Cycles

1.1 One Evaporator - SC1

The schematic for the One Evaporator Subcritical Heat Pump cycle (SC1) is shown in Figure 1.

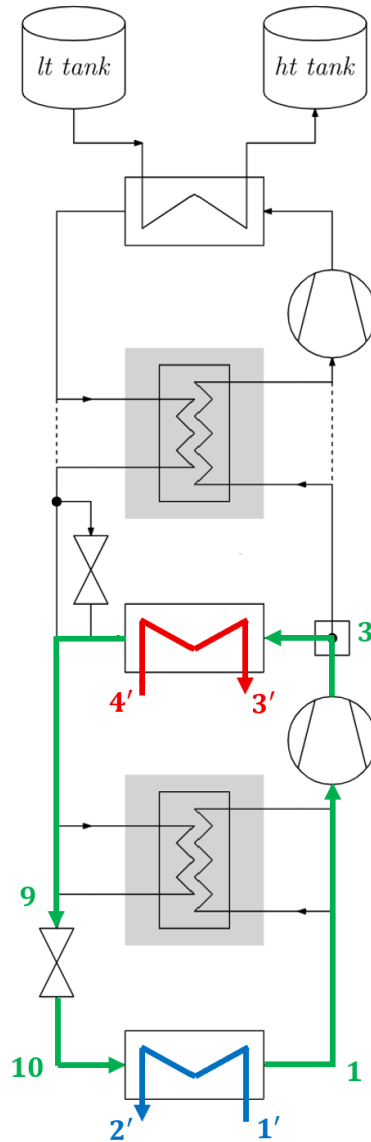


Figure 1: Schematic of the One Evaporator Subcritical Heat Pump cycle (SC1)

To be completed

1.2 One Evaporator with Recuperator - SC1R

The schematic for the One Evaporator Subcritical Heat Pump cycle with Recuperator (SC1R) is shown in Figure 2.

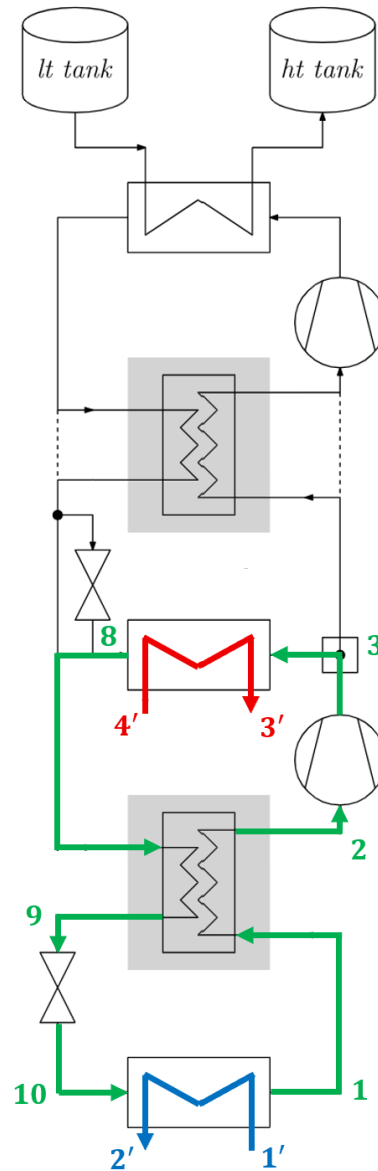


Figure 2: Schematic of the One Evaporator Subcritical Heat Pump cycle with Recuperator (SC1R)

To be completed

1.3 Two Evaporators - SC2

The schematic for the Dual-Evaporator Subcritical Heat Pump cycle (SC2) is shown in Figure 3.

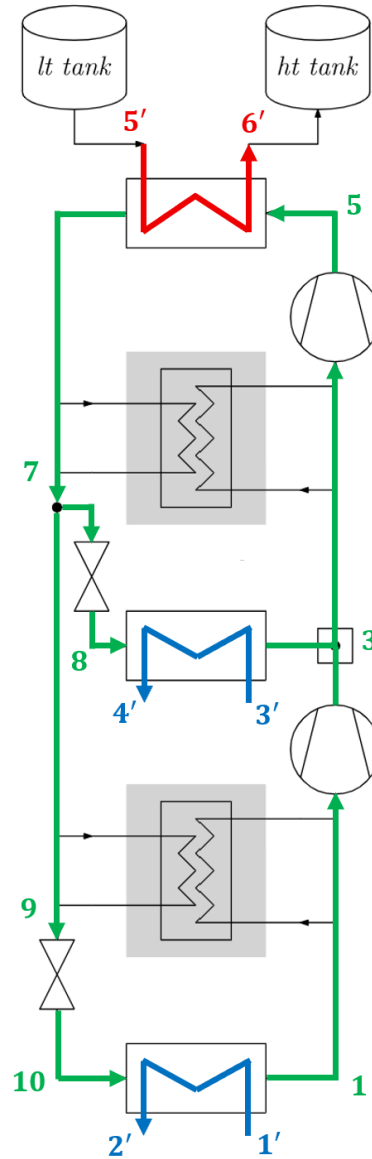


Figure 3: Schematic of the Dual-Evaporator Subcritical Heat Pump cycle (SC2)

To be completed

1.4 Two Evaporators with Recuperators - SC2R

The schematic for the Dual-Evaporator Subcritical Heat Pump cycle with Recuperators (SC2R) is shown in Figure 4.

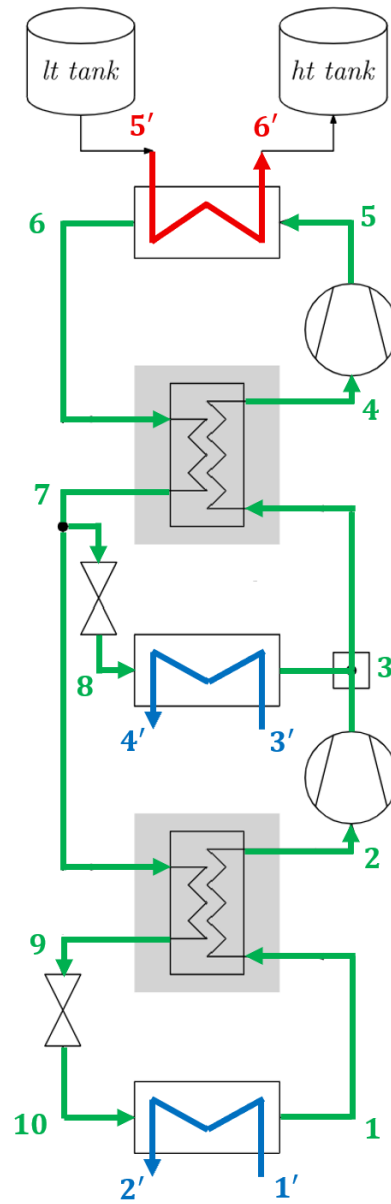


Figure 4: Schematic of the Dual-Evaporator Subcritical Heat Pump cycle with Recuperator (SC2R)

To be completed

2 Transcritical Cycles

2.1 One Evaporator - TC1

The schematic for the One Evaporator Transcritical Heat Pump cycle (TC1) is shown in Figure 5.

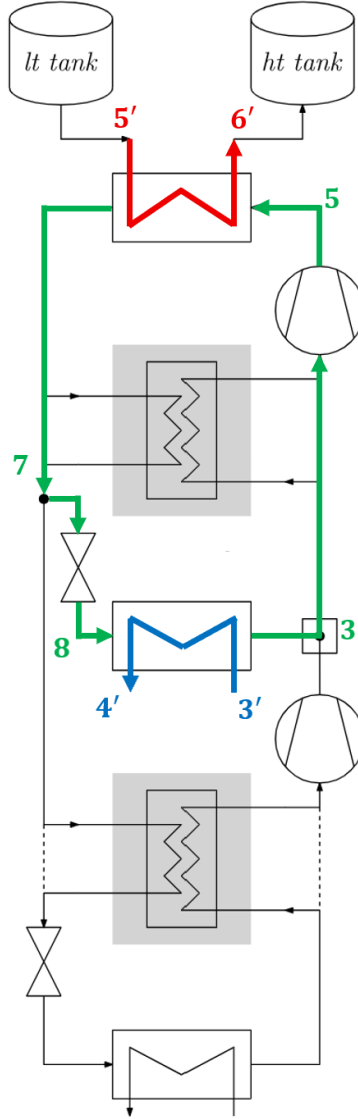


Figure 5: Schematic of the One Evaporator Transcritical Heat Pump cycle (TC1)

To be completed

2.2 One Evaporator with Recuperator - TC1R

The schematic for the One Evaporator Transcritical Heat Pump cycle with Recuperator (TC1R) is shown in Figure 6.

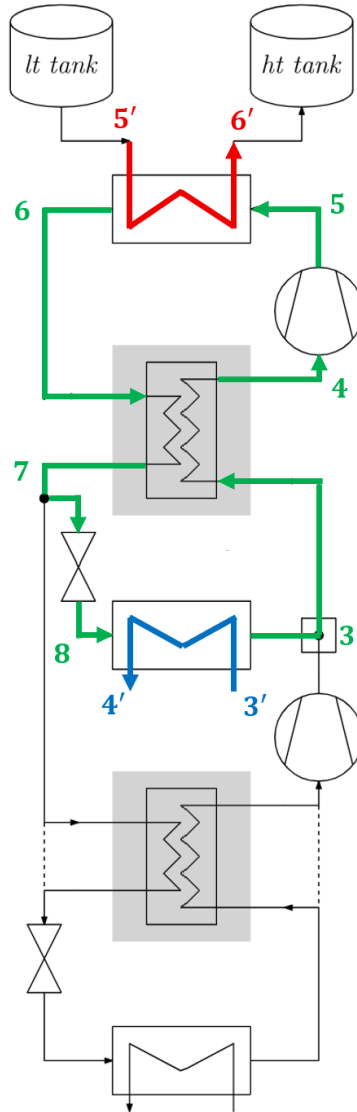


Figure 6: Schematic of the One Evaporator Transcritical Heat Pump cycle with Recuperator (TC1R)

To be completed

2.3 Two Evaporators - TC2

The schematic for the Dual-Evaporator Transcritical Heat Pump cycle (TC2) is shown in Figure 7.

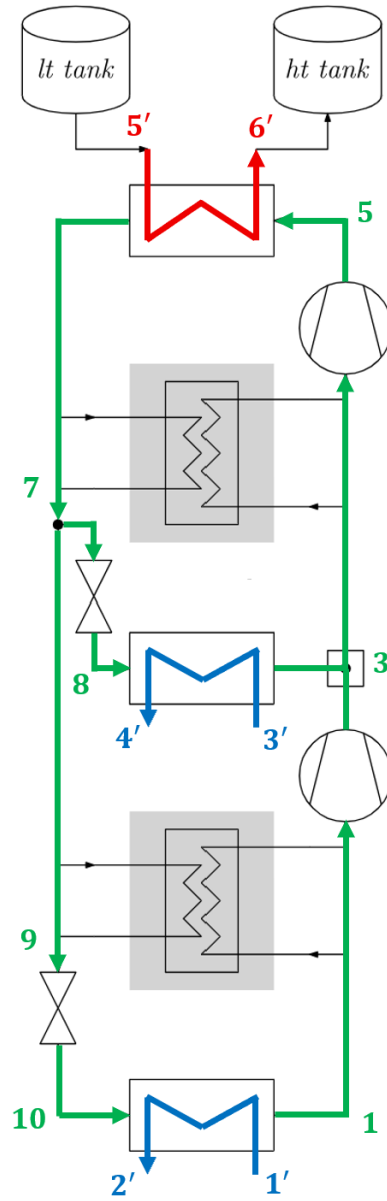


Figure 7: Schematic of the Dual-Evaporator Transcritical Heat Pump cycle (TC2)

To be completed

2.4 Two Evaporators with Recuperators - TC2R

The schematic for the Dual-Evaporator Transcritical Heat Pump cycle with Recuperators (TC2R) is shown in Figure 8.

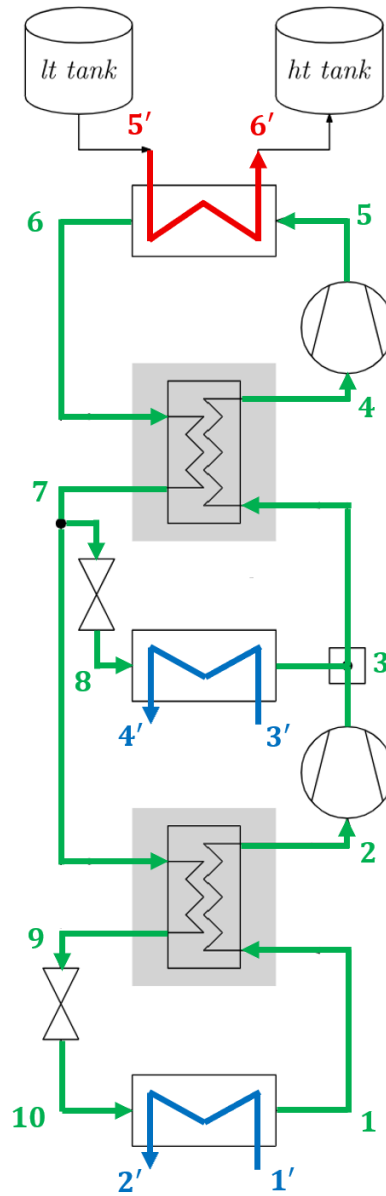


Figure 8: Schematic of the Dual-Evaporator Transcritical Heat Pump cycle with Recuperator (TC2R)

To be completed